

Package ‘hdcrt’

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Type Package

Title Hypothesis testing in high-dimensional censored-transformation models

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Description We provide an efficient censored rank-based test statistic for hypothesis testing in high-dimensional censored-transformation models. Both global test and partial test are supported.

License GPL (>= 2)

Imports Matrix

Repository github

Encoding UTF-8

LazyData true

LazyDataCompression xz

URL <https://github.com/XiaoZhangryy/hscrt>

BugReports <https://github.com/XiaoZhangryy/hscrt/issues>

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NeedsCompilation yes

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hdcrypt

*High dimensional censored rank partial test***Description**

High dimensional censored rank partial test

Usage

```
hdcrypt(
  x,
  y,
  status,
  z,
  smooth = c("sigmoid", "gaussian"),
  h = NULL,
  covariate_dependence = TRUE
)
```

Arguments

x	The design matrix.
y	The survival outcome.
status	The right censoring indicator.
z	The control vector, which is the control factors multiplied by the estimated coefficients.
smooth	The smooth function used. "sigmoid" represents the sigmoid prime kernel function and "gaussian" represents the gaussian kernel function.
h	The bandwidth of the control vector. The default value is the standard deviation of z divided by the square root of sample size n.
covariate_dependence	An indicator of whether the censoring mechanism is dependent on covariates.

Value

A list.

- ts - Test statistic.
- pval - p value.

See Also

[hdcrt](#), [hdcrt_dc](#)

Examples

```
set.seed(0)
n <- 150
p <- 550
x <- matrix(rnorm(n * p), n, p)
u <- matrix(rnorm(n * p), n, p)
alpha <- c(rep(1, 5), rep(0, p - 5))
beta <- c(rep(1, 5), rep(0, p - 5))
y <- u %*% alpha + x %*% beta + rnorm(n)
status <- sample(c(0, 1), n, replace = TRUE, prob = c(0.2, 0.8))
fit <- sprfabs(y, u, status)
alphahat <- fit$opttheta
z <- u %*% alphahat
test_result_sigmoid <- hdcrt(x, y, status, z, "sigmoid")
print(test_result_sigmoid)
test_result_gaussian <- hdcrt(x, y, status, z, "gaussian")
print(test_result_gaussian)
```

hdcrt

High dimensional censored rank test

Description

High dimensional censored rank test

Usage

```
hdcrt(x, y, status, covariate_dependence = TRUE)
```

Arguments

x	The design matrix.
y	The survival outcome.
status	The right censoring indicator.
covariate_dependence	An indicator of whether the censoring mechanism is dependent on covariates.

Value

A list.

- ts - Test statistic.
- pval - p value.

See Also

[hdcrt](#), [hdcrt_dc](#)

Examples

```

set.seed(0)
n <- 150
p <- 550
x <- matrix(rnorm(n * p), n, p)
beta <- c(rep(1, 5), rep(0, p - 5))
y <- x %*% beta + rnorm(n)
status <- sample(c(0, 1), n, replace = TRUE, prob = c(0.2, 0.8))
test_result <- hdcrt(x, y, status)
print(test_result)

```

hdcrt_dc	<i>High dimensional censored rank test with independent double censored</i>
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Description

High dimensional censored rank test with independent double censored

Usage

```
hdcrt_dc(x, y, status_left, status_right)
```

Arguments

x	The design matrix.
y	The survival outcome.
status_left	The left censoring indicator, independent of x.
status_right	The right censoring indicator, independent of x.

Value

A list.

- ts - Test statistic.
- pval - p value.

See Also

[hdcrtpt](#), [hdcrt](#)

Examples

```

set.seed(0)
n <- 150
p <- 550
x <- matrix(rnorm(n * p), n, p)
beta <- c(rep(1, 5), rep(0, p - 5))
y <- x %*% beta + rnorm(n)
status_left <- y >= qnorm(0.1) * sqrt(6)
status_right <- y <= qnorm(0.9) * sqrt(6)
test_result <- hdcrt_dc(x, y, status_left, status_right)
print(test_result)

```

sprfabs	<i>A forward and backward stagewise algorithm for high-dimensional spr problem.</i>
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Description

A forward and backward stagewise algorithm for high-dimensional spr problem.

Usage

```
sprfabs(
  y,
  x,
  status,
  eps = 0.01,
  xi = 1e-10,
  maxIter = 3000,
  sigma = NULL,
  weight = NULL,
  nmax = NULL,
  lam_m = NULL,
  message = TRUE
)
```

Arguments

y	The response.
x	The design matrix.
status	The right censoring indicator.
eps	The step size for updating coefficients. Default is $\text{eps} = 0.01$.
xi	The threshold for qfabs. Default is $\text{xi} = 1e-10$.
maxIter	The maximum number of outer-loop iterations allowed. Default is $\text{maxIter} = 3000$.
sigma	The tuning parameter in the Sigmoid function. Default is NULL.
weight	An optional weights. Default is 1 for each observation.
nmax	Limit the maximum number of variables in the model. When exceed this limit, program will early stopped. Default is NULL.
lam_m	The ratio of the minumum lambda and the maximum lambda. Default is NULL.
message	An indicator of whether print warning messages.

Value

A list.

- theta - The estimation of covariates.
- opttheta - The optimal estimation of covariates.
- lambda - lambda sequence.

- direction - direction sequence.
- iter - Iterations.
- bic - The EBIC for each solution.
- loss - loss for each solution.
- df - Number of nonzero coefficients.
- opt - Position of the optimal lambda based on EBIC.

Examples

```
set.seed(0)
n <- 150
p <- 550
x <- matrix(rnorm(n * p), n, p)
alpha <- c(rep(1, 5), rep(0, p - 5))
y <- x %*% alpha + rnorm(n)
status <- sample(c(0, 1), n, replace = TRUE, prob = c(0.2, 0.8))
fit <- sprfabs(y, x, status)
alphahat <- as.vector(fit$opttheta)
print(which(alphahat != 0))
print(alphahat[alphahat != 0])
```

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